

DISASTRACE: DEVELOPER & TECHNICAL DEFENSE Q&A

Complete answers to Conceptual Framework, Non-functional Specs, Architecture, and Security.

1. CONCEPTUAL FRAMEWORK

1. Ano po yung mga ginamit na software or software requirements?

- **Web Front-End Framework:** Next.js (React 19, App Router)
- **Mobile Front-End Framework:** React Native with Expo (SDK 52, Expo Router)
- **Styling Engine:** Tailwind CSS v4 (Web) & NativeWind / Tailwind CSS (Mobile)
- **Programming Language:** TypeScript (100% strict type safety)
- **Backend Runtime / API:** Next.js API Routes (Node.js Serverless runtime)
- **Database & BaaS Platform:** Supabase (PostgreSQL with PostGIS spatial extension)
- **Database ORM:** Drizzle ORM (type-safe SQL builder)
- **Mapping & Map Tile Engine:** MapLibre GL (@maplibre/maplibre-react-native for Mobile, OpenFreeMap / Leaflet for Web)
- **PDF Export Engine:** jsPDF & jsPDF-AutoTable with HTML5 Canvas vector-to-PNG renderer
- **Real-time Engine:** Supabase Realtime (WebSockets & Postgres CDC Channels)
- **UI Components & Icons:** Shadcn UI, Radix UI, Lucide React, Lucide React Native, Iconsax

2. NON-FUNCTIONAL REQUIREMENTS (PORTABILITY)

1. Saan po portable/compatible na operating system ang mobile front-end ng system/app? Yung web-based naman po, saang search engine (web browser) naman po siya masesearch/maaaccess?

- **Mobile App Compatibility (Operating Systems):**
 - Android: Android 8.0 (Oreo / API Level 26) at mas mataas (Android 9, 10, 11, 12, 13, 14, 15).
 - iOS: iOS 14.0 at mas mataas (iPhone 8 at mga bagong modelo).
- **Web Dashboard Compatibility (Web Browsers):**
 - Ang web dashboard ay compatible sa lahat ng modern web browsers na sumusuporta sa HTML5, ES6+, WebSockets, at WebGL:
 - * Google Chrome (v90+)
 - * Microsoft Edge (Chromium-based v90+)
 - * Mozilla Firefox (v88+)
 - * Brave Browser / Opera
 - * Apple Safari (v14+)

3. TECHNICAL FEASIBILITY

1. Saan po nadevelop yung system or ano pong ginamit na platform or IDE? Pati po yung database na ginamit.

- **Development Platform / IDE:** Visual Studio Code (VS Code) / Antigravity IDE
- **Version Control & Repository:** Git & GitHub
- **Deployment / Cloud Platforms:**
 - Web Application & API Backend: Vercel Cloud Platform
 - Database & Authentication Services: Supabase Cloud (Managed Infrastructure)
 - Mobile Build Engine: Expo Application Services (EAS Build / Android APK)

4. COMPATIBILITY CHECKING

1. Device Platform, 2. Software Versions, 3. CPU/Storage (Ano po yung certain units na compatible?):

A. Resident & Responder Mobile App Specs:

- Device Platform: Android Smartphone o Apple iPhone
- Minimum Software Version: Android 8.0 (Oreo) o iOS 14.0
- Recommended Software Version: Android 11+ o iOS 16+
- Minimum CPU: Quad-Core 1.5 GHz Processor
- Recommended CPU: Octa-Core 2.0 GHz Processor o mas mabilis
- Minimum RAM: 2 GB RAM (3 GB+ recommended)
- Storage Space: 100 MB na libreng storage space para sa app at offline draft caching (disas_trace_drafts.json)
- Hardware Sensors Required: GPS / Location Sensor, Internet Connectivity (Wi-Fi o 3G/4G/5G Cellular Data), Camera (para sa incident photo evidence)

B. PACC Dispatcher & CDRRMO Admin Web Dashboard Specs:

- Device Platform: Desktop Computer, Laptop, o Workstation Tablet
- Operating System: Windows 10/11, macOS 11+, Linux (Ubuntu/Debian)
- Recommended Web Browser: Google Chrome, Microsoft Edge, o Brave (Latest Version)
- Minimum Hardware: Dual-Core 2.0 GHz CPU, 4 GB RAM, Full HD Display (1920x1080 resolution)

5. TRACKING

1. Google Maps API po ba yung ginamit sa system for tracking?

- Sagot: Hindi po Google Maps API. Ang system ay gumagamit ng OpenFreeMap / MapLibre GL kasama ang OpenStreetMap (OSM) tiles at Expo Location Services.
- Bakit ito ang ginamit?
 1. Zero Cost / No Paywall: Hindi kailangan magbayad ng mahal na API billing katulad sa Google Maps API.
 2. High-Performance Map Rendering: Gumagamit ng vector tiles at WebGL/Native rendering sa mobile (@maplibre/maplibre-react-native).
 3. Real-time Live Telemetry: Ang GPS location tracking ay tumatakbo gamit ang Expo Location (watchPositionAsync) at isino-broadcast sa PACC Dashboard sa pamanamagitan ng Supabase Realtime WebSockets at PostGIS spatial indexing (locationGeom).

6. ECONOMIC FEASIBILITY

1. Baka po may masama po kayo sa expenses na nagastos.

- Web Hosting & API Deployment (Vercel): P0.00 / free (Free Tier during dev)
- Database, Auth & Realtime (Supabase Cloud): P0.00 / free (Free Tier up to 500MB DB & 50k MAU)
- Map Tiles & Geocoding (OpenFreeMap / OSM): P0.00 / free (Open-source alternative to Google Maps)
- Mobile App Build / Testing (Expo CLI / EAS): P0.00 / free
- Development Tools & IDE (VS Code / Antigravity IDE): P0.00 / free
- Domain Name (Optional): ~P600.00 - P800.00 / taon
- KABUOANG GASTOS (DEV PHASE): P0.00 (Zero-cost open-source stack)

7. CYBERSECURITY & SYSTEM SECURITY

[Authentication & Authorization]

- **How are user passwords stored? (e.g., hashed using bcrypt or Argon2):**

- Ang mga password ay hindi kailanman nakasulat nang plain text. Ang authentication ay pinamamahalaan ng Supabase Auth na gumagamit ng malakas na Argon2id (o bcrypt sa pamamagitan ng PostgreSQL pgcrypto at Supabase GoTrue engine) na may kasamang cryptographic salt.

- **Does the system implement Role-Based Access Control (RBAC)?**

- Oo. Mayroong apat (4) na malinaw na antas ng role: pacc_admin (PACC Dispatchers), cdrrmo_admin (CDRRMO Executives), ambulance_responder (Emergency Crew), at resident / user (General Public).

- **Can users only access features based on their assigned role?**

- Oo. Ipinapatupad ito sa Front-end Navigation Guards (useAuthStatus) at Back-end Server Guards (/api routes) na nagso-validate ng session Bearer token.

- **Is there a session timeout for inactive users?**

- Oo. Ang Supabase Auth JWT access tokens ay may awtomatikong expiry (1 hour). Pagkatapos nito, nangangailangan ito ng valid refresh token.

[Database Security]

- **How do we prevent SQL Injection attacks?**

- Ang system ay 100% protektado laban sa SQL Injection dahil hindi ito gumagamit ng unescaped string concatenation sa mga database query.

- **Are prepared statements or parameterized queries used?**

- Oo. Lahat ng database queries ay dumadaan sa Drizzle ORM at Supabase Client SDK, na naggagarantiyang gumamit ng parameterized queries / prepared statements.

- **Is database access limited only to authorized users or services?**

- Oo. Ang database ay protektado ng Row Level Security (RLS) sa Supabase PostgreSQL. Ang database connection string ay nakakubli sa server-side environment variables.

- **Are regular database backups implemented?**

- Oo. Ang Supabase Cloud backend ay may awtomatikong Daily Database Backups at WAL (Write-Ahead Logging) Point-In-Time Recovery (PITR).

[Incident Reporting Security]

- **How do we prevent fake or spam incident reports?**

- Required Live GPS Location, mandatory camera photo evidence, at authenticated user account validation.

- **Is there a verification process before dispatching responders?**

- Oo. Lahat ng emergency request ay dumadaan muna sa PACC Verification Queue (VERIFICATION_PENDING) bago i-verify (VERIFIED) at irelihi ang responder.

- **Are all incident reports logged for auditing?**

- Oo. Lahat ng aksyon (mula SOS report, PACC verification, dispatch offer, acceptance, decline, telemetry logs, hanggang PCR at Trip Ticket) ay awtomatikong naitatala sa audit tables.

[Availability & Disaster Recovery]

- **What happens if the server goes down?**

- Serverless Resilience sa Vercel at Supabase Cloud. Kapag nagkaroon ng outage, ang mobile app ay may Offline Caching & Queue System (use-offline-reports.ts).

- **Is there a backup and recovery plan?**

- Oo. Ang natapos na PCR at Trip Ticket ay nase-save sa local storage ng cellphone (disas_trace_drafts.json) habang offline, at may 5-Minute Recurring System Reminder na nag-aalerto sa responder kapag nagkaroon uli ng internet connection.

- **How often are backups performed?**

- Automated daily full database snapshots at continuous Write-Ahead Logging (WAL) replication.

[General Security Practices]

- **How are software updates managed?**

- Git version control at CI/CD deployment pipelines sa Vercel.

- **Are security patches applied regularly?**

- Oo. Regular na ina-update ang packages at sinusuri gamit ang npm audit.

- **Has the system been tested for vulnerabilities?**

- Oo. Na-test ang system laban sa SQL Injection & XSS (Zod & Drizzle ORM), Unauthorized Route Access (RBAC), at Offline Data Integrity.

- **What cybersecurity standards or best practices are followed?**

- OWASP Top 10 Security Guidelines, HTTPS / TLS 1.3 Encryption, REST API Bearer Token Authentication (JWT), at Environment Variables Secrecy.